

NXR-1700RE

VHF Analog & Digital Base Station

NXDN™ **DMR**

MULTI-MODE, SPACE SAVING SOLUTION

The Kenwood NXR-1700RE Base station continues the line of migration friendly and forward thinking infrastructure to offer the ultimate in modern scalability and security for analog, NXDN™ and DMR two-way systems in a new and compact form-factor. Within a built-in antenna relay for simplex or duplex operation, the NXR-1700RE offers a reliable and space-saving solution for networking wide area analog and digital voice and data calls, enhanced secure remote programming, support for SSL certificates, SNMP monitoring, event and hardware logging, as well as tiered user management roles.



Features

- VHF 136-174 50 – 1 W RF Output Power (Up to 50 W @ 50%, 25 W @ 100% Duty Cycle)
- Non-repeat Simplex / Semi-Duplex Mode for Analog and NXDN Digital and DMR Tier II*2
- Light, Compact and Space-Efficient
- 1.71-inch OLED with Icons and Numeric Displays
- Thermal-Controlled Cooling Fan
- Up to 32 Channels
- Selectable Digital Protocol: DMR Tier II / NXDN Conventional*2 (programmable one at a time)
- USB-A Connector for Audio Accessories
- External In/Out Pin from DB25
- Hot Standby System Redundancy
- Built-in IP Network Adapter
- Unicast and Multicast Call Routing
- SNMP protocol for network management
- Supports G.711 Audio Codec (for Test console and Third-party Applications)
- IP Remote Management (Monitor / Control / Programming / Test Console)
- IPIF to External Applications (for IP Console, OTAP)
- IP Network Voice Logging Interface
- Built-in SIP IF (Digital Only)*1,*2
- IP Remote Control Interface (IPRCI)
- Enhanced Security (HTTPS)
- CW ID
- Multi-Site Conventional IP Network up to 16 Sites (for both Digital and Analog)*2
- IP Networking Compatible with NXR-5000 (NXDN)*2, legacy KTI-3 (NXDN)*2 and KTI-5 (DMR)*2

Digital – Common

Built-in AMBE+2™ Vocoder	Encrypted Voice/Data (AES / DES / DMR Enhanced Encryption)
Mixed Analog / Digital Operation	User List / Site Group Table
Site Roaming with Beacon	

Digital – NXDN™ Mode

FDMA – Very narrow 6.25 kHz & narrow 12.5 kHz bandwidths	NXDN Digital Site Roaming
NXDN Conventional Operation*2	Radio Access Control*1

Digital – DMR Mode

TDMA 2-slot 12.5 kHz bandwidth equivalent to 6.25 kHz very narrow bandwidth	DMR Site Roaming
DMR Tier II Conventional Operation*2	Call Interruption

FM Modes – General

FM Conventional Operation	Analog call routing
Multiple QT/DQT	Voting

*1 Requires version upgrade of terminal to obtain compatibility with this model

*2 Software option

Supplied Accessories

DC Power Cable

Single mounting bracket

Dual mounting bracket

Rack handles

Example of rack mount options.

Specifications

General		NX1P-1700RE	
Frequency Range		136 - 174 MHz	
Type 1			
Type 2			
Channel Capacity		32	
Channel Spacing			
Analog		30* / 25* / 15 / 12.5 kHz	
Digital		12.5 / 6.25 kHz	
PLL Channel Step		2.5 / 3.125 / 5 / 6.25 kHz	
Power Supply		10.8 - 15.6 V DC	
Current Drain		0.6 A	
Standby		12.0 A (Max. power), 9.0 A (25 W)	
Transmitting			
Operating Temperature		-22 °F to + 140 °F (-30 °C to +60 °C)	
Frequency Stability		± 0.5 ppm	
Antenna Impedance		50 Ω	
Dimensions		(W x H x D) Projections Not Included 8.21 x 1.73 x 8.33 in (208.5 x 44.0 x 211.5 mm)	
Weight Radio		4.18 lb (1.9 kg)	
FCC ID			
Type 1		K44513100	
ISED Certification			
Type 1		282F-513100	
*25/30 kHz in VHF/UHF Bands excluding T-Band are not included in the models sold in the USA or US territories.			
Analog measurements made per TIA603. Specifications are measured according to applicable standards. Specifications shown are typical and subject to change without notice, due to advancements in technology. Details and timing of firmware and software updates are subject to change without notice.			

Receiver		NXR-1700RE	
Sensitivity			
DMR (5 % BER)		0.22 µV	
DMR (1 % BER)		0.28 µV	
NXDN 12.5 / 6.25 kHz (3% BER)		0.25 / 0.20 µV	
Analog (12dB SINAD)		0.25 µV	
Selectivity			
Analog @ 25 / 12.5kHz (TIA603)		83 / 77 dB	
Analog @ 25 / 12.5kHz (TIA603E)		80 / 50 dB	
FM Hum & Noise			
Analog @ 12.5kHz		50 dB	
Analog @ 25kHz		55 dB	
Intermodulation		80 dB	
Spurious Rejection		90 dB	

Transmitter		NXR-1700RE	
RF Power Output		50 - 11 W (50 W @ 50% Duty 25 W @ 100 % Duty)	
Spurious Emission		-80 dB	
FM Hum & Noise			
Analog @ 12.5kHz		50 dB	
Analog @ 25kHz		55 dB	
Audio Distortion		1%	
Digital Protocol (DMR)		ETSI TS 102 361-1, -2, -3	
Emission Designator		16K0F3E, 11K0F3E, 7K60FXD, 7K60F7D, 7K60FXE, 7K60F7E, 7K60FXW, 7K60F7W, 8K30F1E, 8K30F1D, 8K30F7W, 4K00F1E, 4K00F1D, 4K00F7W, 4K00F2D	

NXDN™ is a registered trademark of JVCKENWOOD Corporation and Icom Inc.
NEXEDGE® & FleetSync® are a registered trademarks of JVCKENWOOD Corporation.
AMBE+2™ is a trademark of Digital Voice Systems Inc.
All other trademarks are the property of their respective holders.

MIL-STD & IP

MIL Standard	MIL 810F Methods/Procedures	MIL 810G Methods/Procedures	MIL 810H Methods/Procedures
High Temperature	501.4/Procedure I, II	501.5/Procedure I, II	501.7/Procedure I, II
Low Temperature	502.4/Procedure II	502.5/Procedure II	502.7/Procedure II
Temperature Shock	503.4/Procedure I, II	503.5/Procedure I	503.7/Procedure I